Sentinel-1/2 Image Fusion Coupled With CSR, GAN, and Temporal Phenology Feature Construction for Cropland Mapping

Mengqing Pang, *Graduate Student Member, IEEE*, Qihao Chen, *Member, IEEE*, Jiali Shang, Weijia Long, and Xiuguo Liu, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) and optical image fusion can leverage the strength of both data sources for improved ground object identification. However, existing fusion methods often encounter challenges such as spatial information loss and spectral distortion, and few studies address scenarios where fused images struggle to distinguish between distinct ground objects simultaneously, such as crops at different growth stages and low woodland. To address these challenges, we propose a hybrid fusion method for Sentinel-1/2 images and construct a temporal phenology feature for improved crop mapping. To efficiently preserve spatial and spectral information, our method combines principal component analysis (PCA) transform and utilizes convolution sparse representation (CSR) and generative adversarial network (GAN) to fuse the high- and low-frequency coefficients of nonsubsampled shearlet transform (NSST), respectively. Furthermore, leveraging the annual normalized difference vegetation index (NDVI) curve to capture the crop growth cycle, we analyze ground object change patterns across a growth cycle and construct temporal phenology feature based on fused images to enhance crop classification accuracy. Results demonstrate that our method outperforms six classical methods and five state-of-the-art deep learning methods across both subjective and objective evaluation performance. When compared with independent SAR, optical, and fused images, the combination of fused images and temporal phenology features improves the distinction between cropland and woodland and achieves the highest classification accuracy. We also expand the application scenarios to a total of 12 agricultural regions in China, Canada, USA, and France, yielding satisfactory results.**

***Index Terms*— Convolution sparse representation (CSR), crop mapping, image fusion, synthetic aperture radar (SAR), temporal phenology feature.**

I. Introduction

REMOTE sensing satellites offer extensive coverage and frequent revisit intervals, making them valuable for agricultural monitoring [1], [2]. Accurate crop classification mapping is important for monitoring crop growth, estimating yield, controlling pests, and other applications while also supplying essential data for studying global change, as well as the flow of material and energy [3], [4]. Sentinel-1 (S1) and Sentinel-2 (S2) satellites provide free, high-temporal-resolution synthetic aperture radar (SAR) and optical imagery that effectively track changes in crop growth stages, making them ideal data sources for crop classification [5].

Due to constraints of using a single imaging mode, such as limited imaging conditions and narrow applicability, relying solely on one remote sensing data source for Earth observation can be inadequate. Therefore, combining multisource images can be advantageous as it allows us to leverage the complementary information provided by different sources [6], [7], [8]. Certain ground features may appear similar in one type of data, such as cropland and water bodies in optical images, but are more distinct in SAR images. Conversely, distinguishing between buildings and cropland may be difficult in SAR images yet easily discernible in optical images [9]. For this phenomenon, fusing SAR and optical images can integrate characteristics of both to better identify ground objects' distribution [10], [11]. Nevertheless, when both SAR and optical images struggle to distinguish different ground objects, such as when crops enter the heading or flowering stage and the vegetation index reaches its peak, cropland may resemble some low woodland or grassland areas [12]. This highlights the importance of incorporating temporal information when utilizing fused images for land cover classification.

The fused images as the input of classification algorithms can improve the accuracy of crop classification [13]. Thus, choosing an appropriate fusion method to obtain high-quality fused images is essential for achieving accurate classification [14]. Fusion methods can be categorized into three types based on fusion levels: pixel level, feature level, and decision level. Pixel-level fusion methods, offering higher precision and minimal data variation, are particularly suitable for remote sensing applications [15]. Among pixel-level fusion methods, four primary categories exist: component substitution (CS) methods, multiscale decomposition (MSD) methods, hybrid methods, and model-based methods [10]. Hybrid methods, which blend the strengths of CS and MSD, are generally preferred for effectively enhancing spatial and spectral resolution. In recent years, deep learning advancements have significantly impacted image fusion techniques. Deep learning-based image

Received 26 August 2024; revised 4 December 2024; accepted 5 January 2025. Date of publication 9 January 2025; date of current version 28 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 41771467, in part by the Key Research and Development Program of Hubei under Grant 2022BCA080, and in part by Hubei Natural Resources Science and Technology Project under Grant ZRZY2023KJ01. *(Corresponding author: Xiuguo Liu.)*

Mengqing Pang, Qihao Chen, Weijia Long, and Xiuguo Liu are with the School of Geography and Information Engineering, China University of Geosciences, Wuhan 430074, China (e-mail: pmq@cug.edu.cn; chenqihao@cug.edu.cn; 1202210863@cug.edu.cn; liuxg318@163.com).

Jiali Shang is with Ottawa Research and Development Center, Agriculture and Agri-Food Canada, Ottawa, ON K1A 0C6, Canada (e-mail: jiali.shang@agr.gc.ca).

Digital Object Identifier 10.1109/TGRS.2025.3527515

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

fusion technique commonly utilizes convolutional neural network (CNN) models as a foundational framework. Most CNN models are supervised learning, which needs to construct label data [16]. However, due to the complex nature of SAR imaging principle and method, constructing high-quality label data can be challenging. In contrast, the generative adversarial network (GAN) model is the unsupervised deep learning model that can generate a more realistic image without labels, overcoming the limitations of CNN models in SAR and optical image fusion [17], [18], [19]. In addition, image representation methods derived from deep learning, such as convolution sparse representation (CSR), effectively preserve image details with fewer sparse coefficients, proving their effectiveness in image fusion [20], [21].

A vast volume of Earth observation imagery is available daily, with improvements in spatial and temporal resolution, along with advancements in data fusion techniques, enhancing the accuracy of land cover mapping [22]. SAR uses actively emitted microwaves to acquire backscattering energy related to surface roughness, complex dielectric constant, and moisture of objects [23]. However, soil roughness and vegetation growth can pose challenges for the application of SAR images, as SAR signal saturation is caused by varying combinations of soil roughness and leaf area index (LAI) values producing the same signal [24]. There are also some crop spectral similarities on optical images or high soil interference due to lack of growth during the growth stages [25]. Multitemporal images offer an effective solution to these challenges in crop mapping. Using time series information can not only distinguish vegetation and nonvegetation areas but also effectively identify different vegetation types [12], [26]. Studies have found that images taken during peak growth periods are more useful for crop mapping than those taken during low growth periods [27].

Although some progress has been made, challenges remain in the fusion of SAR and optical image for land cover classification.

1) Although hybrid fusion methods offer evident advantages in reducing spatial and spectral distortion, the effectiveness of fusion strategies for multiscale decomposition within these methods significantly impacts the quality of fusion results. Thus, it is imperative to design appropriate fusion techniques tailored to different decomposition coefficients to effectively preserve source image information.
2) When SAR and optical images encounter difficulty in discerning similar ground objects (e.g., crops at heading or flowering stages and low woodland), the potential of fused images becomes constrained. To address the issues of misclassification of ground objects due to crop growth, further research is needed to explore how the characteristics of ground objects in fused images evolve throughout the growth cycle.

To address the outlined issues, our key contributions are as follows.

1) Combined with principal component analysis (PCA) transform, we propose a novel hybrid fusion method of Sentinel-1/2 images in the nonsubsampled shearlet transform (NSST) domain that designs CSR and GAN fusion strategies to produce fused images with rich spatial and spectral information.
2) We construct a temporal phenology feature considering crop growth dynamics and employ the random forest (RF) algorithm for classification, demonstrating the fused images and the constructed temporal phenology feature's ability to enhance the accuracy of land cover classification. In addition, we apply the fused images and temporal phenology feature to a total of 12 agricultural regions in China, Canada, USA, and France, yielding promising results.

II. Methodology

Based on SAR and optical image fusion to improve the discriminability of different ground objects, our approach involves four key steps.

1) *Image Preprocessing:* Sentinel Application Platform (SNAP) software carries out speckle filter, terrain correction, and other preprocessing operations for Sentinel-1 data. The speckle filter employs a refined Lee filter with a size of 5×5. Sentinel-2 Multispectral Instrument (MSI) Level-2A data are obtained through the sen2cor plugin, which performs atmospheric correction on Sentinel-2 MSI Level-1C data. Image registration is carried out automatically using ENVI's module, followed by manual verification to ensure accurate alignment.
2) *Image Fusion:* We propose a hybrid fusion method for Sentinel-1/2 images, integrating CSR and GAN in the NSST domain along with PCA transform. This fusion method is utilized to fuse the preprocessed SAR and optical images to generate fused images.
3) *Feature Construction:* Ground object characteristics are analyzed across the entire crop growth cycle using fused images. Temporal phenology features are constructed based on these fused images during the analysis.
4) *Mapping and Assessment:* An RF algorithm is utilized to map polarized images, optical images, fused images, and the combination of fused images with temporal phenology features. This enables the evaluation of the recognition effectiveness of fused images and temporal phenology features for crops and woodland at different growth stages. The detailed steps are shown in Fig. 1.

A. Sentinel-1/2 Image Fusion

The detailed fusion process includes the following steps: 1) PCA transform processes S2 optical images to extract the first principal component (PC1) and other principal components (PCs); 2) histogram matching is then performed on PC1 and S1 SAR images; 3) the matched PC1 and SAR images are decomposed using NSST to extract the high-frequency and low-frequency coefficients; 4) high-frequency coefficients from PC1 and SAR images are fused using CSR; 5) low-frequency coefficients from PC1 and SAR images are fused using GAN; and 6) the final fused image is obtained by employing inverse NSST and inverse PCA transform.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

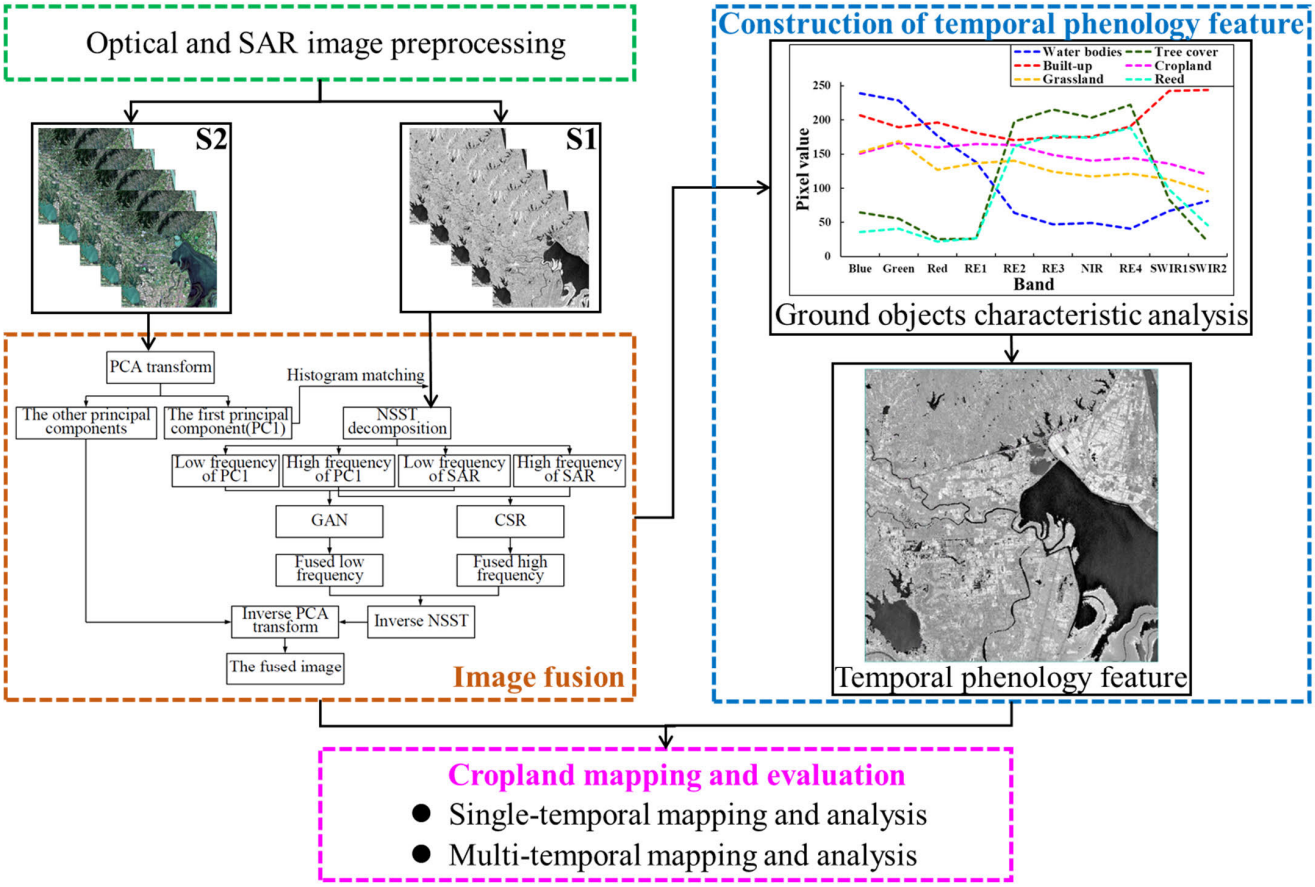


Fig. 1. Overall experimental framework.

1) PCA Transform of S2 Images: PCA is a technique used to eliminate data redundancy while preserving key information [28]. By deriving eigenvalues and eigenvectors from the correlation matrix, it transforms the original correlated dataset into an equal number of orthogonal and uncorrelated PCs. Among these, the PC1 has the highest variance, integrating the greatest amount of information from the source image [29]. In our method, we apply PCA transform to the optical image to obtain PC1 and other PCs. Then, PC1 and SAR images are matched and fused to obtain a new PC1. Finally, an inverse PCA transform is applied to the new PC1 and other PCs to generate the final fused image.

2) NSST Decomposition of PC1 and S1 Images: NSST, initially introduced by Easley et al. [30], stands out as one of the most advanced multiscale decomposition methods to date. We begin by performing histogram matching on PC1 and SAR images, followed by L-level decomposition, yielding $L+1$ subbands comprising one low-frequency subband and L high-frequency subbands. Subsequently, the high-frequency subbands from each level of multiscale decomposition undergo further processing using a shift-invariant shearlet filter bank (SFB) in a multidirectional manner. Fig. 2 shows the decomposition process of a two-level NSST.

3) CSR High-Frequency Coefficients Fusion: High-frequency coefficients primarily capture the edge details of the source image, and the goal of high-frequency coefficient fusion is to synthesize the critical features contained within them [31]. The CSR model effectively preserves image details with fewer sparse coefficients, making it theoretically well suited for high-frequency coefficient fusion. Here, we introduce the CSR model into the fusion of high-frequency coefficients from multiscale decomposition, specifically designed to handle the high-frequency components of PC1 and SAR image decomposition.

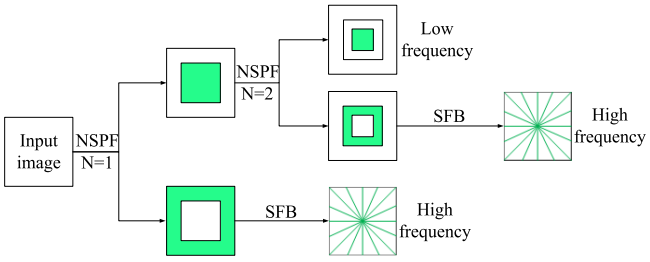


Fig. 2. Schematic of two-level NSST decomposition.

CSR represents the source image S as the sum of a set of convolutions between sparse coefficient maps x_m and dictionary filters d_m, which are the same size as the image [21], [32]

$$\arg\min_{\{\mathrm{x}_m\}} \frac{1}{2}\left\|\sum_m \mathrm{d}_m * \mathrm{x}_m - \mathrm{S}\right\|_2^2 + \lambda \sum_m \|\mathrm{x}_m\|_1 \qquad (1)$$

where $*$ denotes the convolution operation. Suppose that there are K pre-registered source images, denoted as $\mathrm{S}_k, k \in$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

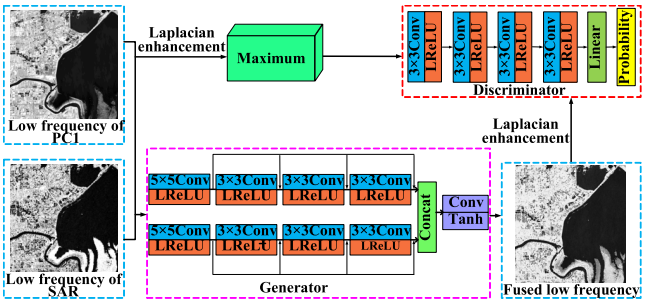


Fig. 3. GAN low-frequency fusion strategy.

$\{1, \ldots, K\}$, and assume that a set of dictionary filters $\mathrm{d}_m, m \in \{1, \ldots, M\}$ has been learned by the dictionary learning method in [33]. S_k^h represents the high-frequency coefficient obtained by S_k image decomposition, with sparse coefficient maps $C_m^k, m \in \{1, \ldots, M\}$, which can be obtained using the following equation:

$$\arg\min_{\{C_m^k\}} \frac{1}{2}\left\|\sum_{m=1}^{M} \mathrm{d}_m * C_m^k - \mathrm{S}_k^h\right\|_2^2 + \lambda \sum_{m=1}^{M}\left\|C_m^k\right\|_1 \tag{2}$$

where $C_{1:M}^k(x, y)$ represents the content of position (x, y), and it is obvious that $C_{1:M}^k(x, y)$ is an M-dimensional vector. Then, the "choose-max" strategy is applied to achieve the fused coefficient maps

$$C_{1:M}^F(x, y) = \begin{cases} C_{1:M}^A(x, y), & \text{if } \left|C_{1:M}^A(x, y)\right|_1 \geq \left|C_{1:M}^B(x, y)\right|_1 \\ C_{1:M}^B(x, y), & \text{otherwise.} \end{cases} \tag{3}$$

We reconstruct the fused sparse coefficients to acquire fused high-frequency subbands that meet the expected requirements. The reconstruction rule is given in the following formula:

$$H = \sum_{m=1}^{M} \mathrm{d}_m * C_m^F. \tag{4}$$

4) GAN Low-Frequency Coefficient Fusion: Drawing from the MFF-GAN model [18], we design the GAN low-frequency fusion strategy, as shown in Fig. 3. The Laplacian operator is applied for generating gradient maps for low-frequency coefficients. Among them, the generator uses two same paths for information extraction and reuses features based on DenseNet [34]. The convolution kernel sizes of the first layer and other three layers are 5×5 and 3×3, respectively. The fused images are produced by a 1×1 convolution layer. The discriminator comprises four 3×3 convolution layers and a linear layer.

The loss functions comprise two parts: generator loss $\mathcal{L}_G$ and discriminator loss $\mathcal{L}_D$.

The generator loss $\mathcal{L}_G$ is composed of content loss C and adversarial loss A. m serves the purpose of balancing and adjusting the two losses. The specific formulas are as follows:

$$\mathcal{L}_G = A + mC \tag{5}$$

$$A = \frac{1}{N}\sum_{n=1}^{N}\left(D\left(\nabla\left(I_f^n\right)\right) - a\right)^2 \tag{6}$$

where N stands for the total number of fused images, $D(\cdot)$ stands for the discriminator function, a is the probability value of discriminator to recognize false data as real data, and $\nabla(\cdot)$ stands for using the Laplacian operator to generate the corresponding gradient maps.

Content loss C includes intensity loss $\mathcal{L}_i$ and gradient loss $\mathcal{L}_g$. The purpose of $\mu(\cdot)$ is utilized for balancing and adjusting the intensity and gradient losses. The formula is as follows:

$$C = \mu_1 \mathcal{L}_i + \mu_2 \mathcal{L}_g \tag{7}$$

$$\mathcal{L}_i = \frac{1}{HW}\left(\|I_f - H_A\|_F^2 + \|I_f - H_B\|_F^2\right) \tag{8}$$

$$\mathcal{L}_g = \frac{1}{HW}\left(\|\nabla I_f - \nabla H_A\|^2 + \|\nabla I_f - \nabla H_B\|^2\right) \tag{9}$$

where H stands for height; W stands for width; H_A and H_B correspond to the source data A and B, respectively; and I_f represents the fusion result.

The loss function of discriminator is $\mathcal{L}_D$. The gradient map is formalized as G_f and the joint gradient map is formalized as G_{joint}, where abs$(\cdot)$ and max$(\cdot)$ stand for the absolute value function and maximum function, respectively. The specific formulas are as follows:

$$G_f = \text{abs}\left(\nabla I_f\right) \tag{10}$$

$$G_{\text{joint}} = \max(\text{abs}(\nabla I_1), \text{abs}(\nabla I_2)) \tag{11}$$

$$\mathcal{L}_D = \frac{1}{N}\sum_{n=1}^{N}\left[D\left(G_f^n\right) - b\right]^2 + \left[D\left(G_{\text{joint}}^n\right) - c\right]^2 \tag{12}$$

where b and c are the labels of the gradient map and joint gradient map, respectively. b should be set to 0, and c should be set to 1.

B. Constructing Feature Based on Fused Images

The integration of SAR and optical images provides complementary information about a given area, thereby enhancing the accuracy of land cover classification. In instances where distinguishing between land cover types proves challenging in either SAR or optical images alone, image fusion emerges as a potent tool for enhancing ground object identification. In scenarios where both SAR and optical images face difficulties in distinguishing certain land cover types, such as during crop heading or flowering stage and some low woodland areas, we need to analyze ground object characteristics within fused images and construct the feature considering time series information to optimize the monitoring and discriminatory capability of the fused images for ground classes.

1) Analysis of Ground Object Characteristics: The normalized difference vegetation index (NDVI) curves for six ground objects in 2021 are presented in Fig. 4(a). Based on these NDVI curves, the growth cycle of the crops was determined, including the period when the crop entered the heading or flowering stage. Analysis indicates that the crop growth cycle in this region spans from early June to mid-October, with the heading or flowering period from late July to mid-August. For characteristic analysis of ground objects, fused

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

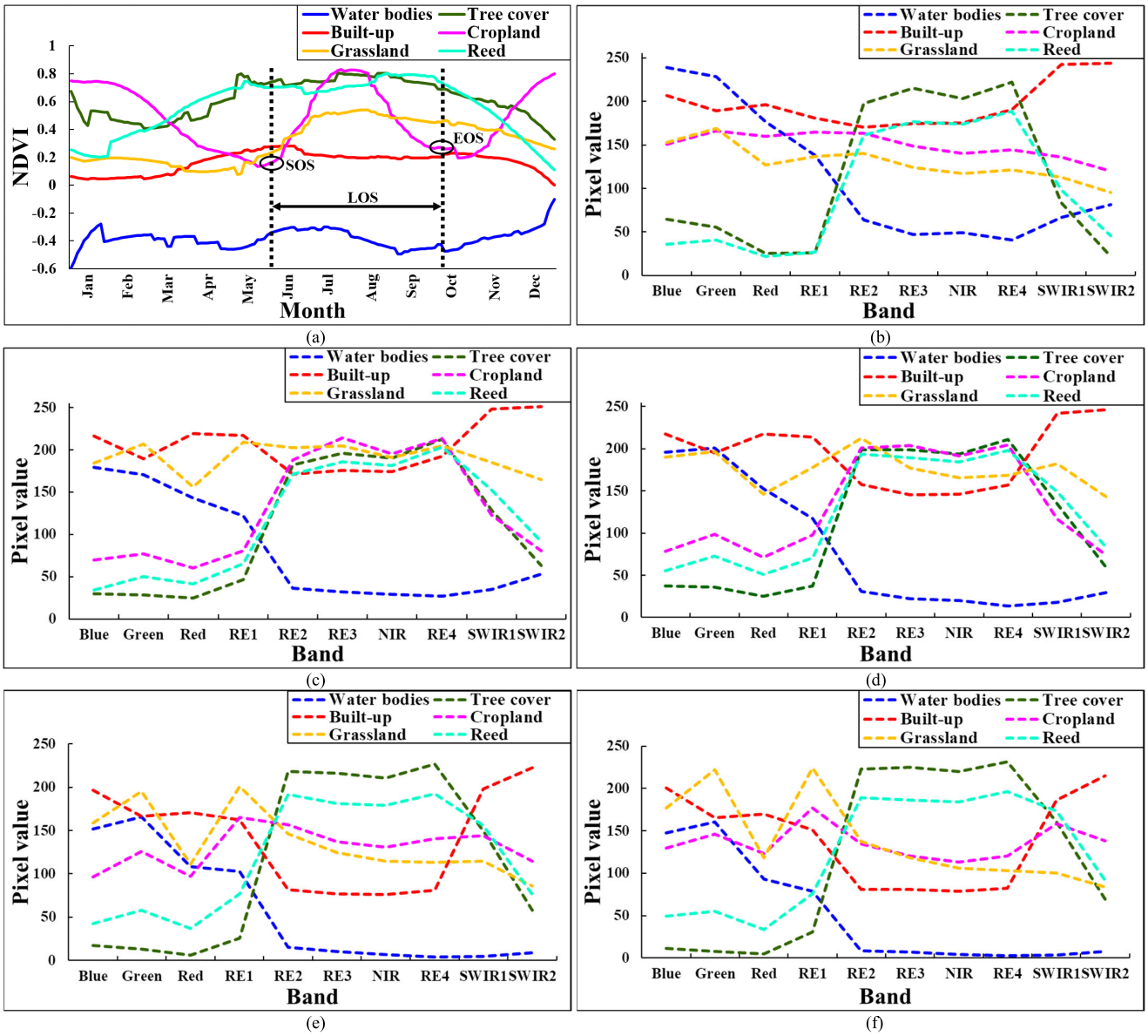


Fig. 4. Characteristic analysis of ground objects. (a) NDVI curves of six ground objects in 2021. (b)–(f) Grayscale value curves of the six ground objects on the fused images from June to October.

images from a growth cycle (June–October) were selected. Fig. 4(b)–(f) illustrates the gray value curves of six ground objects on the fused images from June to October. Among them, RE1–RE4 correspond to four red-edge (RE) bands, while SWIR1 and SWIR2 represent two shortwave infrared (SWIR) bands. The fused images from July and August reveal that cropland peaks in the RE2, RE3, near-infrared (NIR), and RE4 bands, closely resemble the gray value of woodland during this period, while the two are well distinguished in other months. This similarity during the heading or flowering stage of crops suggests the higher misclassification between cropland and woodland.

2) Construction of Temporal Phenology Feature: Based on the analysis of aforementioned ground object characteristics and acknowledging the superior clarity of texture features in the NIR band of Sentinel-2 imagery, which offers a 10-m resolution, we developed a temporal phenology feature ($\mathrm{FI_NIR}_t$) using the NIR band from fused images captured during the early growth and heading or flowering stages of crops. The specific construction rules are detailed as follows:

$$\mathrm{FI_NIR}_t = \frac{\mathrm{FI_NIR}_{tm} - \mathrm{FI_NIR}_{ts}}{\mathrm{FI_NIR}_{tm} + \mathrm{FI_NIR}_{ts}} \tag{13}$$

where $\mathrm{FI_NIR}_{tm}$ represents the NIR band of the fused images when the crop enters the heading or flowering stage, corresponding to the peak NDVI values, and $\mathrm{FI_NIR}_{ts}$ represents the NIR band of the fused images during the early sowing period. The value of $\mathrm{FI_NIR}_t$ is dimensionless and ranges from -1 to 1.

The timing of crop planting and harvesting varies greatly across different regions due to diverse climatic conditions. In our experimental area, crops are typically planted in June and reach the heading or flowering stage between late July and mid-August. Consequently, $\mathrm{FI_NIR}_{tm}$ corresponds to the NIR band of the fused image from August, while $\mathrm{FI_NIR}_{ts}$ corresponds to the NIR band of the fused image from June.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

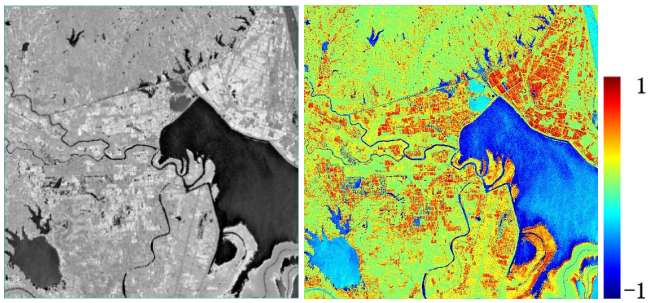


Fig. 5. Temporal phenology feature image. (a) Grayscale image display and (b) false-color image display.

The temporal phenology feature image FI_NIR$_t$, constructed according to formula (13), is shown in Fig. 5.

C. Single-Temporal/Multitemporal Mapping and Analysis

We utilize the high-performing RF machine learning algorithm for cropland mapping. To assess the effectiveness of the fused images and FI_NIR$_t$ feature in identifying ground objects, we conduct single-temporal and multitemporal classification for the fused images and the combination of fusing images and the FI_NIR$_t$ feature.

For single-temporal classification, we designed three feature datasets: 1) Sentinel-2 spectral bands; 2) fused bands; and 3) fused bands plus FI_NIR$_t$ feature. These datasets were used to evaluate the discriminative capability of fused images and the additional FI_NIR$_t$ feature across different land cover types. The addition of the FI_NIR$_t$ feature was intended to assess its impact on the accuracy of single-temporal mapping throughout the entire growth cycle. It is particularly relevant when access to viable images is limited during certain growth stages, and it explores whether satisfactory results can be achieved by combining images from a specific time point with the FI_NIR$_t$ feature, especially during crop stages such as heading or flowering, when images from this period are more affected by clouds and rain and less available.

For multitemporal classification, we designed four feature datasets: 1) Sentinel-1 polarized channels over three months; 2) Sentinel-2 spectral bands over three months; 3) fused bands over three months; and 4) fused bands plus the FI_NIR$_t$ feature over three months. These datasets were used to investigate the effect of fused images and the FI_NIR$_t$ feature on classification accuracy. We evaluated the results utilizing overall accuracy (OA), Kappa coefficient, user's accuracy (UA), producer's accuracy (PA), and the $F1$ score. The $F1$ score offers a comprehensive measure of PA and UA, with a score closer to 1 indicating better the classification performance. The calculation formula is as follows:

$$F1 = (2 * \mathrm{PA} * \mathrm{UA})/(\mathrm{PA} + \mathrm{UA}). \tag{14}$$

III. Experiments and Analysis

A. Study Area and Datasets

Sentinel-1 images utilize VV and VH polarization data acquired on October 5, 2021. Sentinel-2 images utilize four bands at 10-m resolution and six bands at 20-m resolution acquired on October 4, 2021. The temporal gap between the acquisition dates of Sentinel-1 and Sentinel-2 data is only one day. The experimental area encompasses diverse land cover types, including water bodies, buildings, forest, cropland, reed, grassland, and other categories. Fig. 6 shows images of Dongting Lake and its surrounding experimental area, along with some photographs of typical ground objects captured in situ. To assess the effectiveness of fused bands and FI_NIR$_t$ feature, we analyze different scenarios from 12 agricultural regions across four countries on three continents globally, as shown in Fig. 20.

Through on-site investigations and analysis of high-resolution images from Google Earth, supplemented by data from the Dynamic World product and the annual WorldCover product, the experimental area was classified into six distinct land cover categories: water bodies, forest, buildings, cropland, reed, and grassland. The specifics of the selected sample set are detailed in Table I.

TABLE I
Ground Objects Training and Verification Sample Set in the Experimental Area

Class	Training data/pixels	Verifying data/pixels	Total/pixels
Water bodies	13993	49092	63085
Forest	6482	14034	20516
Buildings	3969	9936	13905
Cropland	9348	22390	31738
Grassland	673	3776	4449
Reed	5897	26506	32403

B. Utility Analysis of Fusion Method

This section provides a comprehensive analysis of our method from five key perspectives. First, we introduce the evaluation metrics used for image fusion. Subsequently, we compare our approach with six classical methods: PCA [35], GS [36], NSST-PAPCNN [37], NSCT-SR [38], SR [38], and CSR [32]. In addition, we conduct a comparison with five state-of-the-art deep learning methods: FusionGAN [39], MFF-GAN [18], YDTR [40], SwinFusion [41], and CDDFuse [42]. Third, we use the Canny algorithm to detect image edges on the fused images corresponding to 10- and 20-m resolution bands of the original S2 images to further validate the preservation of spatial details. Fourth, we assess the method's capacity to retain spectral information by calculating the spectral distortion index D_λ and analyzing spectral similarity within spectral profiles. Finally, we analyze the fusion effect of different NSST decomposition levels, and the calculation time and efficiency.

1) Evaluation Metrics: Qualitative evaluation mainly relies on human visual perception, which can introduce subjectivity and variability. To achieve more objective assessment, quantitative metrics are necessary. We selected seven metrics for this purpose: Erreur Relative Globale Adimensionnelle de Synthese (ERGAS) [43], universal image quality index (UIQI) [44], spectral angle mapper (SAM) [45], standard deviation (STD) [46], spatial frequency (SF) [47], root-mean-square error (RMSE) [48], and peak signal-to-noise

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

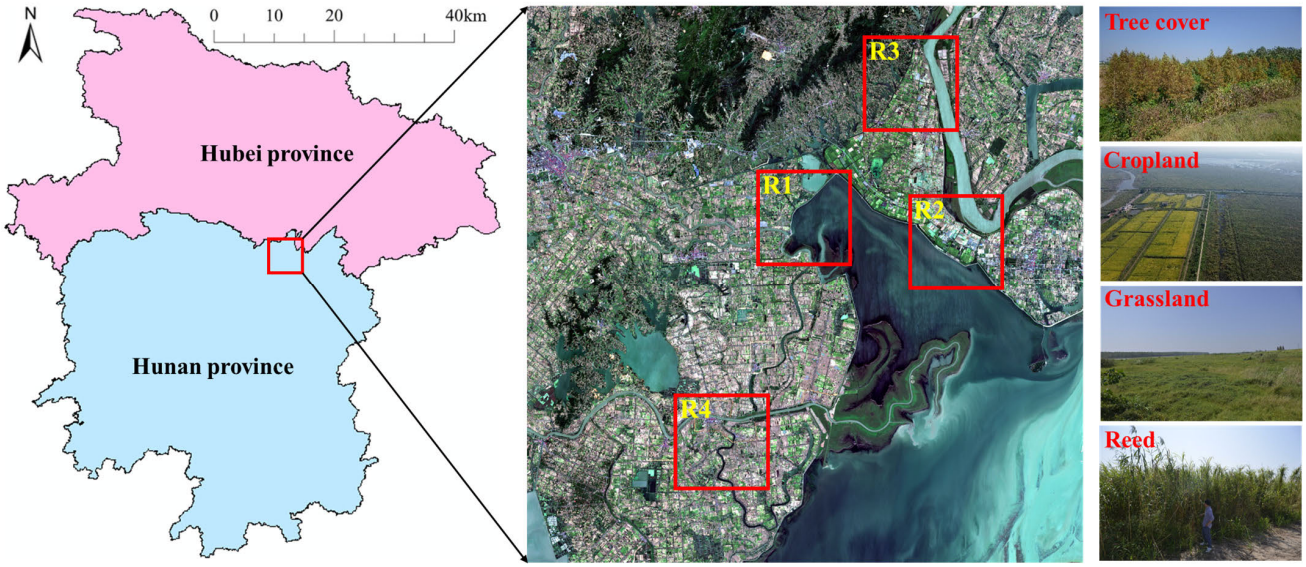


Fig. 6. Images of the experimental area (true color display of red, green, and blue bands) and photographs of typical ground objects.

TABLE II
DESCRIPTION OF QUANTITATIVE METRICS

Metrics	Equation	Description
ERGAS	$100 \times \frac{h}{l} \times \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(\frac{RMSE_k}{\mu_k}\right)^2}$	Represents the overall quality of fusion results. the lower ERGAS values, the higher the fusion quality.
UIQI	$\frac{\sigma_{XF}}{\sigma_X\sigma_F} \cdot \frac{2\mu_X\mu_F}{(\mu_X^2+\mu_F^2)} \cdot \frac{2\sigma_X\sigma_F}{(\sigma_X^2+\sigma_F^2)}$	Represents the distortion degree of image brightness component. The higher UIQI values, the better the brightness.
SAM	$\mathrm{SAM}(x,y) \triangleq \arccos\left(\frac{\langle x,y\rangle}{\|x\|_2 \cdot \|y\|_2}\right)$	Represents the spectral distortion degree. The lower the values, the better quality of fusion results.
STD	$\sqrt{\frac{1}{m\times n}\sum_{i=1}^{m}\sum_{j=1}^{n}(F(i,j)-\mu)^2}$	Represents the degree of dispersion in image gray values concerning the mean gray value. The bigger the values, the richer information of images.
SF	$\sqrt{RF^2+CF^2}$ $RF = \sqrt{\frac{1}{m\times n}\sum_{i=1}^{m}\sum_{j=1}^{n}\lvert F(i,j)-F(i,j-1)\rvert^2}$ $CF = \sqrt{\frac{1}{m\times n}\sum_{i=1}^{m}\sum_{j=1}^{n}\lvert F(i,j)-F(i-1,j)\rvert^2}$	Represents the change rate of image grayscale. The bigger the values, the clearer images, and the better the fusion quality.
RMSE	$\sqrt{\frac{1}{m\times n}\sum_{i=1}^{m}\sum_{i=1}^{n}(F(i,j)-X(i,j))^2}$	Represents the deviation between fused images and source images. The lower the values, the better the effect.
PSNR	$10lg\left\lvert\frac{G_{max}^2}{\frac{1}{m\times n}\sum_{i=1}^{m}\sum_{i=1}^{n}\left(F(i,j)-X(i,j)\right)^2}\right\rvert$	The higher PSNR values are, the stronger the correlation between two images, indicating better fusion quality.

ratio (PSNR) [48]. Table II provides a detailed description of these seven metrics.

2) Comparison of Different Methods:

a) Comparison with classical methods: Fig. 7 presents the fused images generated using six different methods. PCA and GS methods successfully retain the spectral characteristics of the optical images. However, these linear transform-based methods fall short in capturing the spatial and texture details from SAR images, especially in areas with complex surface features. This limitation arises because both PCA and GS primarily focus on spectral fidelity, leading to a loss of spatial details that are critical for accurately representing features in SAR images. The NSST-PAPCNN method performs extensive feature extraction from SAR images, thereby integrating more spatial details. However, the method causes considerable spectral distortion due to its focus on spatial features, which makes spectral discrimination of various objects more challenging in fused images. NSCT-SR and SR methods allow partial retention of spectral information while incorporating SAR details, yet they yield fusion results marked by numerous patches. This patchy effect often arises from sparse representation's reliance on local neighborhood information, which can fragment

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

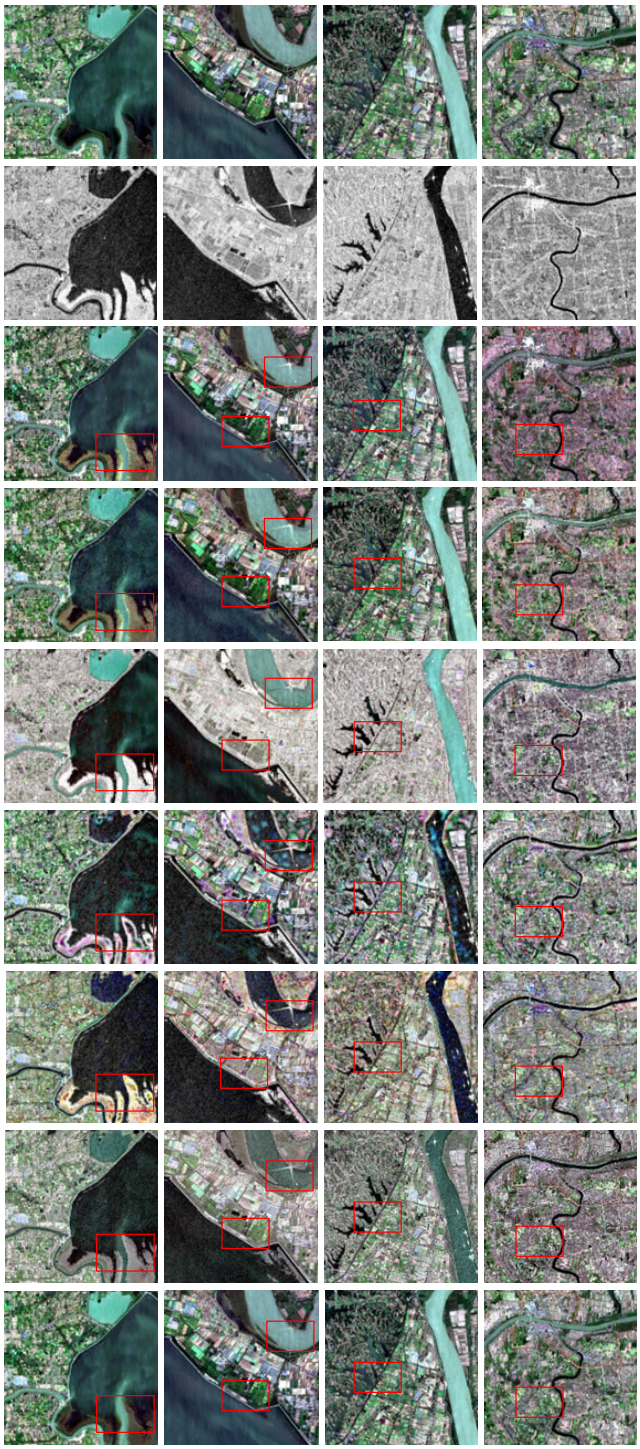


Fig. 7. Results of our method and six classical methods in four regions (each row from top to bottom represents optical image, SAR image, PCA, GS, NSST-PAPCNN, NSCT-SR, SR, CSR, and our method).

continuous features and limit the visual integrity of the fused images. The CSR method effectively integrates SAR spatial information, but it somewhat disrupts spectral consistency. The focus on spatial details in CSR may overshadow the spectral features of the optical images, resulting in inconsistencies in the spectrum between ground objects enhanced with spatial information and those in the optical images. In contrast, our method combines the advantages of PCA, NSST, GAN, and CSR, overcoming the limitations of the aforementioned methods, and generates fused images with richer spatial details and ground object representations that are closer to the true surface.

The fusion results are quantitatively evaluated. Table III displays the average values derived from ten bands of fused images, with optimal results heightened in bold. Our method consistently exhibits superior performance across various metrics, including ERGAS, UIQI, SAM, STD, SF, RMSE, and PSNR. Among them, the ERGAS and SAM scores indicate minimal spectral distortion, validating the method's ability to preserve the spectral characteristics of optical images. The STD, SF, RMSE, and PSNR values demonstrate improvements in image details and contrast. UIQI is a comprehensive metric that indicates the method's good balance in maintaining both spatial and spectral features. In summary, the quantitative results demonstrate the superiority of our method in preserving spatial details and reducing spectral distortion.

b) Comparison with state-of-the-art deep learning methods: Fig. 8 presents the fused images generated by various methods. According to the analysis in Fig. 8, it is evident that the FusionGAN method integrates a significant amount of SAR information, enhancing the texture details in the fused images. However, the recognizability of different ground objects is poor. This issue is particularly noticeable in areas such as cropland around water bodies, which appear uniformly gray, showing a significant deviation from optical images. This indicates that the FusionGAN generative adversarial structure focuses on spatial features while ignoring the preservation of spectral information. The MFF-GAN, YDTR, SwinFusion, and CDDfuse methods effectively integrate SAR spatial information while maintaining a certain degree of spectral features, representing improvements in fusion results compared to FusionGAN. However, these methods still exhibit limitations in spectral differentiation across various land cover types, such as cropland, bare soil, and buildings. In particular, YDTR, SwinFusion, and CDDfuse struggle to maintain distinct spectral features within heterogeneous land cover types. This suggests that while these methods succeed in enhancing spatial detail, their fusion mechanisms are insufficient for fully capturing the complex spectral distributions essential for accurate land cover differentiation. In contrast, our method generates images with rich spatial and spectral information through a balanced fusion of spectral and spatial features.

The quantitative results in Table IV indicate that our method achieves optimal performance across most metrics, including ERGAS, UIQI, SAM, RMSE, and PSNR. The STD metric reflects the degree of dispersion in image gray values relative to the average gray value, with higher values indicating richer image information. Although the SwinFusion method achieves a slight improvement in STD, likely due to its emphasis on SAR spatial textures. Overall, our method outperforms other methods in quantitative evaluation and effectively overcomes the limitations of other methods, providing a fusion solution with high spectral fidelity and rich spatial details.

3) Edge Features Extraction: To further assess the effectiveness of maintaining spatial information, we used the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE RESULTS OF OUR METHOD AND SIX CLASSICAL METHODS IN FOUR REGIONS (AVERAGE VALUES OF TEN BANDS)

Images	Metrics	PCA	GS	NSST-PAPCNN	NSCT-SR	SR	CSR	Our method
Region 1	ERGAS	69.1965	49.5295	50.7279	46.9391	52.0547	**35.0512**	36.9610
	UIQI	0.6564	0.7637	0.6865	0.7108	0.6791	0.7703	**0.7716**
	SAM	6.9151	11.4373	22.0479	20.7970	22.1549	19.1489	**6.8107**
	STD	65.2569	76.5285	76.1032	72.2319	71.0884	69.2663	**78.8195**
	SF	31.6201	30.5329	27.2896	30.0634	26.7862	26.1114	**36.7091**
	RMSE	54.8747	52.7275	68.8532	52.3674	60.7863	38.5553	**35.6807**
	PSNR	13.9382	14.1402	11.4253	13.8146	12.4765	16.4574	**17.9638**
Region 2	ERGAS	70.2513	39.0953	49.9078	42.8508	47.4619	33.9018	**30.2208**
	UIQI	0.6695	0.8105	0.6504	0.6999	0.6722	0.7242	**0.8129**
	SAM	5.5209	10.0921	22.9959	20.8091	22.3694	19.9251	**3.9672**
	STD	59.8455	77.8073	77.1178	70.8209	71.4615	71.1239	**73.1381**
	SF	29.0492	27.5292	25.1004	28.1729	26.0629	24.3011	**30.5695**
	RMSE	47.3037	42.2461	69.9888	46.9013	55.3381	38.2605	**27.7583**
	PSNR	15.7630	16.3117	11.3437	14.7826	13.3530	16.5847	**20.3216**
Region 3	ERGAS	31.8162	25.8228	40.6045	32.3951	39.3548	27.9952	**17.7945**
	UIQI	0.8860	0.8811	0.7635	0.9089	0.8119	0.8427	**0.9252**
	SAM	4.4429	5.5513	24.9371	20.2697	22.3120	21.2161	**3.3552**
	STD	62.3257	68.5972	56.9266	64.8985	60.1600	57.2340	**69.8106**
	SF	35.1423	34.9707	30.2335	34.6662	30.3041	30.0333	**37.7204**
	RMSE	37.1586	36.6345	72.9875	42.0303	55.9167	40.5928	**25.4808**
	PSNR	19.242	20.14891	11.26766	16.13919	13.45211	16.32148	**22.9247**
Region 4	ERGAS	69.5172	36.8203	32.6178	26.0739	32.4358	24.4626	**20.5041**
	UIQI	0.7918	0.8998	0.8804	0.9392	0.9107	0.9367	**0.9570**
	SAM	3.5982	3.0090	21.4587	18.3942	20.5384	19.2214	**2.6597**
	STD	45.3211	57.0764	46.2343	53.7911	44.4422	43.9898	**56.3656**
	SF	33.5374	**38.4224**	31.9000	34.9900	30.4166	30.3989	38.2840
	RMSE	62.7227	51.4092	55.6662	37.8914	47.6142	35.9086	**30.9322**
	PSNR	12.9369	14.0080	13.5168	16.5950	14.6248	17.0687	**18.9610**



(a) (b) (c) (d) (e) (f)

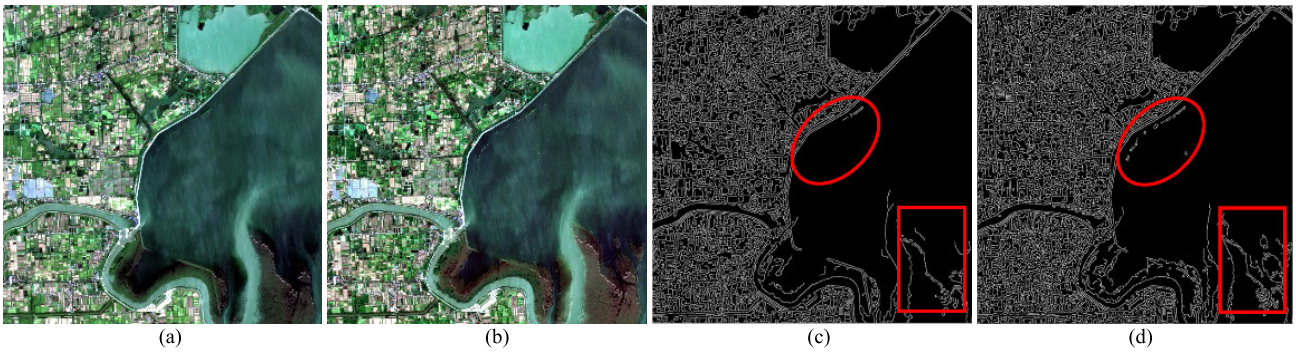
Fig. 8. Fusion results: (a) FusionGAN, (b) MFF-GAN, (c) YDTR, (d) SwinFusion, (e) CDDFuse, and (f) our method.

TABLE IV
QUANTITATIVE RESULTS OF OUR METHOD AND STATE-OF-THE-ART DEEP LEARNING METHODS (AVERAGE VALUES OF TWO REGIONS)

Metrics	ERGAS	UIQI	SAM	STD	SF	RMSE	PSNR
FusionGAN	72.8397	0.6144	25.176	42.5159	15.5079	62.28	12.2948
MFF-GAN	36.7097	0.7132	17.0582	71.2872	29.9314	40.0045	16.1631
YDTR	44.937	0.634	19.717	67.7533	19.5599	63.3622	12.0769
SwinFusion	47.7295	0.6653	20.8619	**85.7056**	25.6705	68.846	11.4199
CDDFuse	46.3622	0.7052	22.6025	82.9325	**34.8501**	60.8962	12.4879
Our method	**33.5909**	**0.7923**	**5.389**	75.9788	33.6393	**31.7195**	**19.1427**

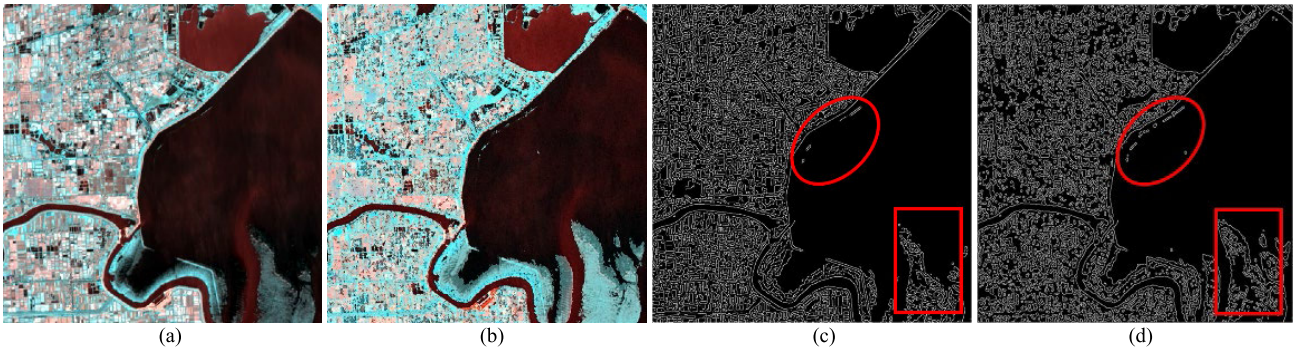
Canny algorithm to extract edge features from both the fused images and optical images. A specific region was selected for experimentation within the aforementioned four local regions. Figs. 9 and 10 depict the extraction results of 10- and 20-m resolution images, respectively. The comparative analysis of the extraction results in Fig. 9 reveals that in contrast to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 9. 10 m resolution images: (a) optical images, (b) fused images, (c) results of optical images edge extraction, and (d) results of fused images edge extraction.



(a) (b) (c) (d)

Fig. 10. 20-m resolution images: (a) optical images, (b) fused images, (c) results of optical images edge extraction, and (d) results of fused images edge extraction.

the optical images, the fused images exhibit better preservation of reed edges around water bodies, accompanied by clearer texture. In addition, other areas show enhanced spatial information. Similarly, the edge extraction results in Fig. 10 indicate that the fused images present richer spatial information and exhibit smoother and more intact edge contours compared with their optical counterparts. In summary, our method demonstrates clear advantages in preserving spatial information.

4) Spectral Quality Analysis:

a) Calculation of spectral distortion index: To evaluate the effectiveness of our method in reducing spectral distortion, we calculated the spectral distortion index D_λ for four local regions [49]. This index quantifies spectral distortion, where a value of 0 denotes optimal performance. Table V demonstrates that our method achieves optimal values across all four regions, indicating that the fused images possess the highest resemblance to optical images concerning spectral features. These findings showcase the remarkable effectiveness of our method in mitigating spectral distortion

$$D_\lambda = P\sqrt{\frac{1}{B(B-1)}\sum_{i=1}^{B}\sum_{j=1,j\neq i}^{B}\left|d_{i,j}(M,F)\right|^P} \tag{15}$$

where $d_{i,j}(M,F) = Q(M_i,M_j) - Q(F_i,F_j)$, which calculates the similarity of spectral features between fused image F and optical image. Q index is UIQI in Table V, and the parameter P is generally set to 1.

b) Analysis of spectral similarity: By comparing changes in gray values at these spectral profile positions, we assess the fused images' capacity to preserve spectral details. Fig. 11 shows the spectral profile positions of the images with 10- and 20-m resolution. Figs. 12 and 13 illustrate gray values along spectral profiles of 10- and 20-m resolution images, respectively. The horizontal axis represents the pixel position along the spectral profile, while the vertical axis displays the corresponding gray values.



(a) (b)

Fig. 11. Spectral profile: (a) 10-m resolution image and (b) 20-m resolution image.

The analysis of Fig. 12 shows that in the S2 optical image, gray values across three bands exhibit clear differences for different ground objects. Specifically, in crop areas, all three bands show high gray values, whereas in water bodies, they show low gray values, with Band 1 notably differing from the other two bands. In fused images generated by the NSST-PAPCNN, NSCT-SR, SR, CSR, and MFF-GAN methods, there is a notable reduction in discrepancies among the bands within the crop areas, with CSR and MFF-GAN methods exhibiting the most pronounced decrease. In addition, the gray values of Band 1 within water bodies tend to converge with those of the other two bands, indicating significant spectral distortion in these fused images. In contrast, fused images obtained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V
RESULTS OF THE SPECTRAL DISTORTION INDEXES OF DIFFERENT METHODS

Regions	PCA	GS	NSST-PAPCNN	NSCT-SR	SR	CSR	MFF-GAN	Our method
Region 1	0.0399	0.0434	0.297	0.3123	0.3579	0.3290	0.3598	**0.0347**
Region 2	0.0494	0.0641	0.3585	0.2908	0.3629	0.3264	0.3548	**0.0384**
Region 3	0.0667	0.0799	0.3235	0.2541	0.3231	0.2974	0.3553	**0.0338**
Region 4	0.0717	0.037	0.2553	0.2313	0.2882	0.2672	0.3132	**0.0246**

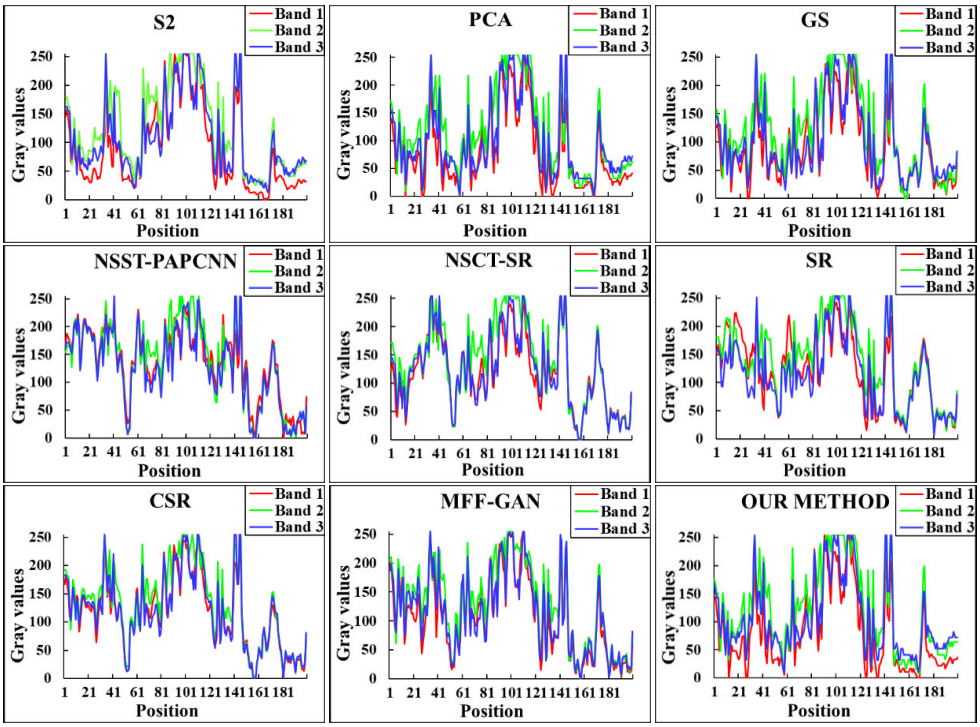


Fig. 12. Extraction results of gray values of 10-m resolution images.

from PCA, GS, and our method preserve spectral information closely resembling that of the S2 optical image, enabling clear differentiation between crop and water bodies areas. This indicates the effectiveness of PCA, GS, and our method in maintaining spectral integrity.

The analysis of Fig. 13 shows that similar to Fig. 12, discernible variations in gray values are noted across the three bands for different ground objects in the S2 image. With the exception of PCA, GS, and our method, all other approaches yield similar gray values across the three bands, suggesting considerable spectral distortion. Conversely, our method efficiently preserves spectral information. Despite the impact of speckle noise in SAR image, which causes significant changes in gray values for some pixels, the overall spectral similarity remains highest.

5) Analysis of Different Decomposition Levels in NSST: Table VI presents the quantitative results for different decomposition levels and corresponding direction numbers in NSST, with the optimal values highlighted in bold. All methods are executed on the same device, the NVIDIA Quadro P6000 GPU. As seen in Table VI, the runtime gradually increases with the increase in decomposition levels and the direction numbers. When the decomposition level is 2, ERGAS, UIQI, SAM, RMSE, and PSNR metrics obtain the optimal values. STD gradually increases with the increase of the decomposition levels, and the optimal value is obtained when the decomposition level is 4. Overall, considering the effect of objective evaluation and the problem of calculation time and efficiency, the decomposition level should be set as 2.

C. Mapping Results and Analysis

1) Single-Temporal Mapping and Analysis: This section evaluates three feature sets: Sentinel-2 spectral bands, fused bands, and fused bands combined with the FI_NIR$_t$ feature. Since constructing temporal phenology features requires images from the peak growth period, the RF algorithm is employed to classify images from the peak growth period and the subsequent months (August–October). The classification results for the three feature sets in August are presented in Fig. 14. The results show relative to the other two feature sets, the combination of fused bands and the FI_NIR$_t$ feature integrates spatial information (from fused bands) and temporal information (from the FI_NIR$_t$), leveraging the dynamic characteristics of crop growth. As a result, it effectively distinguishes between cropland at heading or flowering stages and woodland, reduces salt-and-pepper noise, and maintains land integrity.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

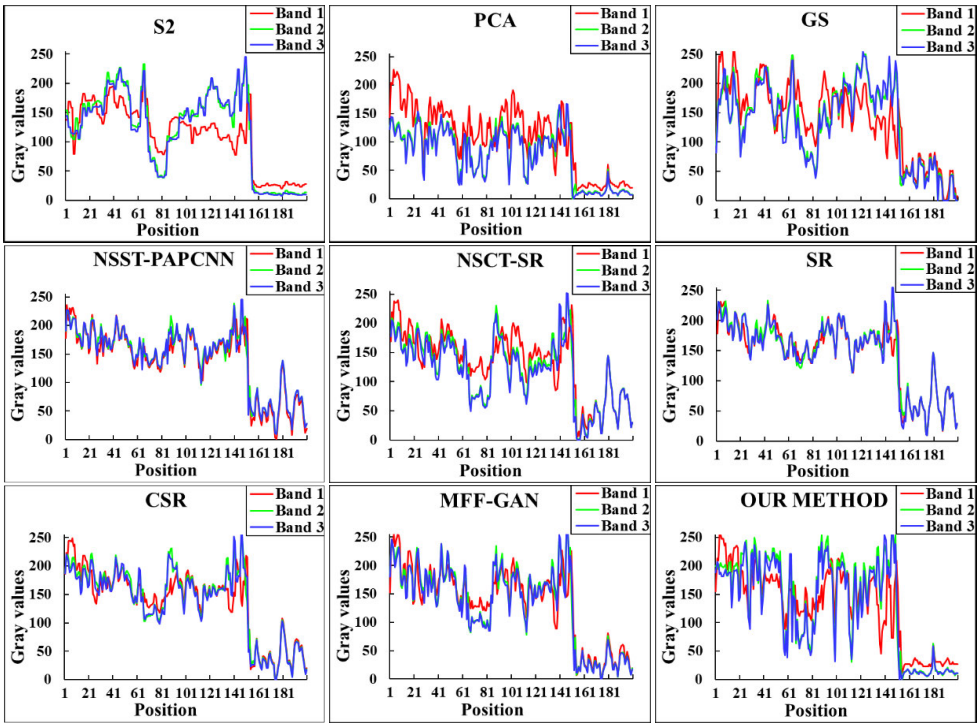


Fig. 13. Extraction results of gray values of 20-m resolution images.

TABLE VI
QUANTITATIVE RESULTS OF DIFFERENT DECOMPOSITION LEVELS AND DIRECTION NUMBERS IN NSST

Metrics	ERGAS	UIQI	SAM	STD	SF	RMSE	PSNR	Time(s)
Level=1, Directions=8	36.0354	0.9276	6.8160	79.1889	**31.2818**	38.8244	16.5551	**24.3414**
Level=1, Directions=16	35.8868	0.9274	5.3890	79.0753	30.1979	38.6695	16.5905	46.7404
Level=2, Directions=16,16	**33.9944**	**0.9345**	**5.1224**	80.7199	28.5205	**37.2957**	**16.8640**	71.0669
Level=3, Directions=16,16,8	34.3967	0.9023	6.6421	84.0996	28.5964	37.8784	16.7259	94.4808
Level=4, Directions=16,16,8,8	34.9974	0.8613	7.2414	**85.8507**	28.5782	38.7087	16.5270	104.5028

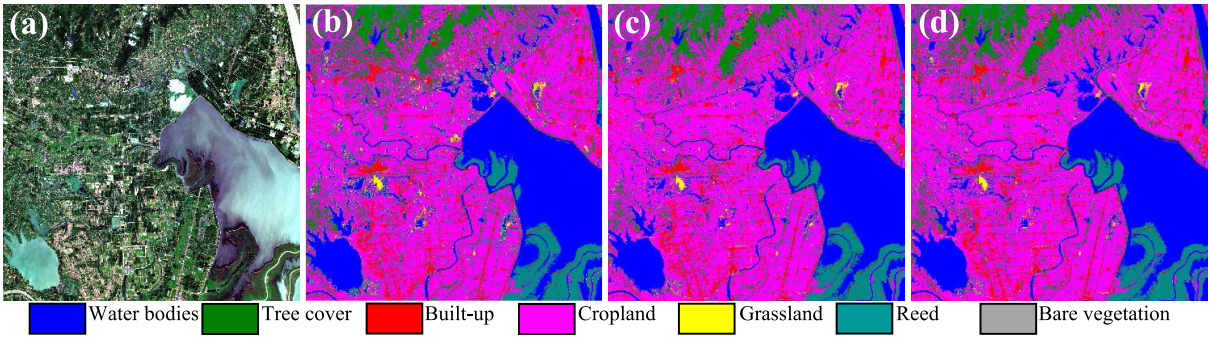


Fig. 14. Classification results in August: (a) optical image, (b) spectral bands, (c) fused bands, and (d) fused bands + FI_NIR$_t$ feature.

We cropped localized areas to provide a more detailed view of the classification results. In Fig. 15, we present the classification results derived from three feature sets over three different months. When only optical bands were used for classification, the land class boundaries became indistinct, accompanied by pronounced salt-and-pepper noise. Notably, cropland frequently underwent misclassified as woodland, and certain regions were misclassified as reed. Such misclassification can be attributed to spectral similarities between woodland and reed during specific months [Fig. 15(a1)–(a3)]. Using fused bands for classification significantly reduced the salt-and-pepper noise and enhanced clarity of land class boundaries, but there was a situation of misclassification of cropland into woodland in some regions [Fig. 15(b2)–(b3)]. Conversely, using fused bands alongside the FI_NIR$_t$ feature yielded the most distinct land class boundaries with minimal salt-and-pepper noise while also improving the differentiation between cropland and woodland [Fig. 15(c1)–(c3)].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

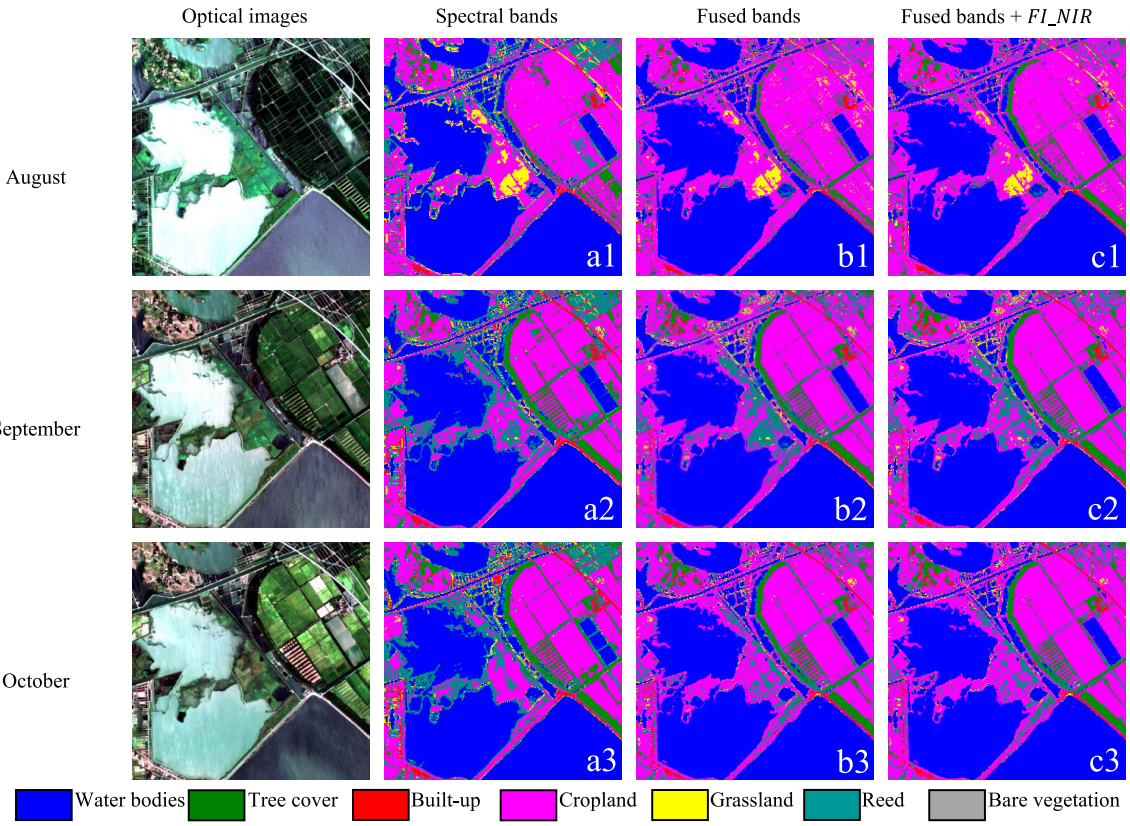


Fig. 15. Classification results of three months.

TABLE VII
QUANTITATIVE EVALUATION OF CLASSIFICATION RESULTS OF DIFFERENT FEATURE SETS

Time	Feature sets		Water bodies	Tree cover	Built-up	Crop-land	Reed	Grass-land	Avg.F1(%)	OA(%)	Kappa
August	Spectral bands	PA	96.36	70.53	78.46	91.90	95.40	62.84	83.81	90.06	0.8673
		UA	97.92	94.87	94.02	85.52	82.06	62.06			
		F1	97.14	80.91	85.54	88.60	88.23	62.45			
	Fused bands	PA	96.25	70.98	89.11	95.92	95.08	56.12	85.28	91.36	0.8846
		UA	97.80	92.51	96.53	88.08	84.43	65.40			
		F1	97.02	80.33	92.67	91.83	89.44	60.40			
	Fused bands+ FI_NIR_t	PA	96.13	76.61	89.45	96.40	95.93	57.23	**86.63**	**92.26**	**0.8968**
		UA	97.92	94.06	96.77	88.93	86.32	67.55			
		F1	97.01	84.44	92.97	92.52	90.87	61.96			
September	Spectral bands	PA	86.91	87.02	79.01	91.12	87.04	50.26	81.04	85.97	0.8142
		UA	94.00	96.00	85.41	84.11	74.34	60.39			
		F1	90.31	91.29	82.09	87.48	80.19	54.86			
	Fused bands	PA	87.98	79.99	79.48	94.75	88.79	56.78	81.99	86.86	0.8258
		UA	95.20	97.38	92.43	82.19	76.60	57.07			
		F1	91.45	87.83	85.47	88.03	82.25	56.92			
	Fused bands+ FI_NIR_t	PA	90.82	82.33	80.92	95.53	89.75	55.56	**83.42**	**88.65**	**0.8492**
		UA	95.82	98.13	93.47	82.54	81.03	59.08			
		F1	93.25	89.54	86.74	88.56	85.17	57.27			
October	Spectral bands	PA	82.36	96.43	71.50	90.60	82.76	59.88	80.65	83.95	0.7882
		UA	91.15	96.51	78.05	83.55	71.94	65.10			
		F1	86.53	96.47	74.63	86.94	76.97	62.38			
	Fused bands	PA	95.62	82.89	73.56	93.68	83.84	62.08	83.40	88.62	0.8478
		UA	94.05	94.36	87.83	79.82	87.04	69.70			
		F1	94.82	88.26	80.06	86.20	85.41	65.67			
	Fused bands+ FI_NIR_t	PA	96.06	87.11	79.58	95.01	85.46	62.42	**85.75**	**90.33**	**0.8708**
		UA	94.91	95.53	91.35	81.51	89.51	73.89			
		F1	95.48	91.13	85.06	87.74	87.44	67.67			

Table VII presents the quantitative evaluation results of three feature sets over three months, with optimal values in bold and suboptimal values underlined. Throughout each month, the combination of fused bands and the FI_NIR_t feature set consistently achieves the highest OA and Kappa. The $F1$ score, serving as a comprehensive metric balancing PA and UA, is analyzed across different ground objects to assess recognition effectiveness. The combination of fused bands

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

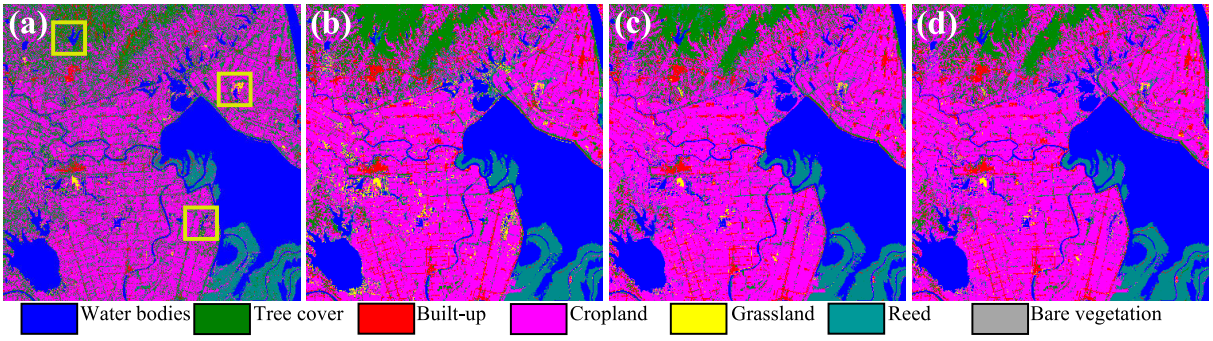


Fig. 16. Temporal classification results: (a) multitemporal polarized channels, (b) multitemporal optical bands, (c) multitemporal fused bands, and (d) multitemporal fused bands combined with the FI_NIR$_t$ feature.

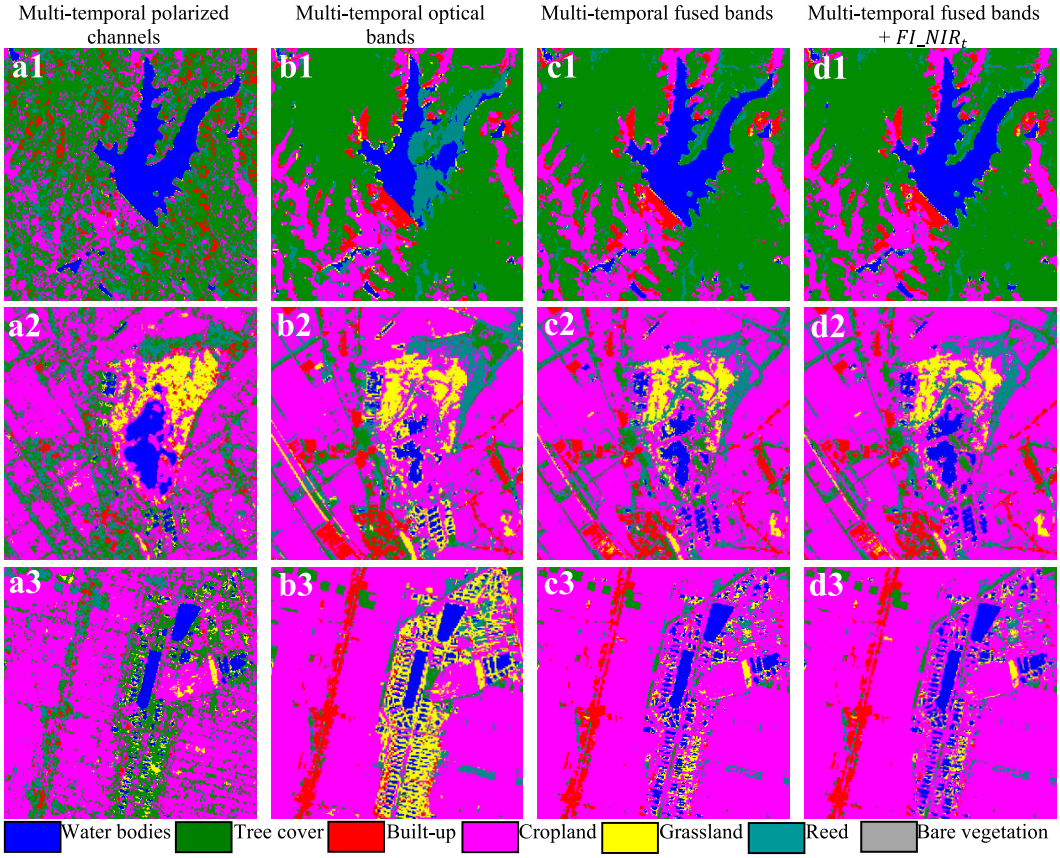


Fig. 17. Temporal classification results for local regions.

and FI_NIR$_t$ feature consistently yield the best or subbest $F1$ values for each type of ground objects, with notable improvements in the accuracy of cropland, buildings, and reed in three months. The accuracy of water bodies and grassland is also highest in September and October. However, the recognition accuracy of grassland is consistently below 80% across all combinations each month. This may be due to the similarity of grassland to the farmland, particularly when mixed within fields, resulting in lower recognition accuracy. Overall, the combination of fused bands and FI_NIR$_t$ feature set effectively enhances the recognition accuracy of various land classes and improves discrimination between them.

2) Multitemporal Mapping and Analysis: This section investigates the influence of fused bands and the FI_NIR$_t$ feature on classification accuracy in multitemporal classification. Four feature sets are utilized for classification: multitemporal polarized channels, multitemporal optical bands, multitemporal fused bands, and multitemporal fused bands in conjunction with the FI_NIR$_t$ feature, as shown in Fig. 16. The results show that the combination of multitemporal fused bands with the FI_NIR$_t$ feature results in the least salt-and-pepper noise, the most distinct land class boundaries, the superior differentiation between land cover types, and yields fewer misclassifications compared with the other three groups.

For a detailed classification results analysis, we cropped specific areas as illustrated in Fig. 17. Employing multitemporal polarized channels for classification resulted in notable misclassifications between cropland and woodland, as well as woodland and buildings. Furthermore, the results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUANTITATIVE EVALUATION OF TIME SERIES CLASSIFICATION RESULTS

Feature sets		Water bodies	Tree cover	Built-up	Cropland	Reed	Grassland	Avg.F1(%)	OA(%)	Kappa
Scheme 1	PA	97.47	47.14	33.38	77.95	70.57	32.28	60.51	75.68	0.6757
	UA	97.76	47.63	39.81	64.38	75.14	47.54			
	F1	**97.62**	47.39	36.31	70.51	72.78	38.45			
Scheme 2	PA	84.68	97.35	84.20	94.37	93.10	64.01	86.30	88.94	0.8543
	UA	96.19	97.23	93.11	89.41	76.34	68.68			
	F1	90.07	**97.29**	88.43	91.83	83.89	66.26			
Scheme 3	PA	96.29	85.36	90.99	98.95	93.72	67.77	89.36	93.73	0.9164
	UA	96.78	98.72	96.76	91.65	90.77	66.06			
	F1	96.54	91.56	93.79	95.16	92.22	66.90			
Scheme 4	PA	96.37	90.51	94.13	99.83	94.58	95.42	**93.95**	**95.75**	**0.9436**
	UA	98.21	98.94	98.34	94.11	93.83	75.98			
	F1	97.28	94.54	**96.19**	**96.89**	**94.21**	**84.60**			

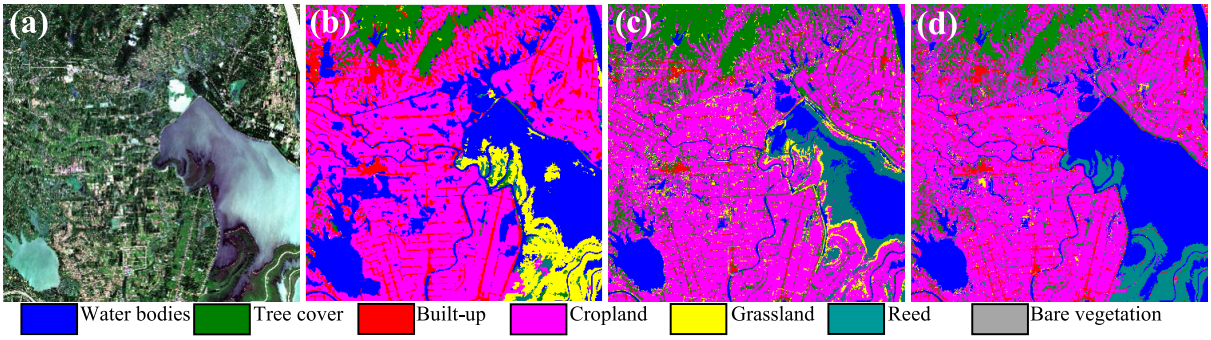


Fig. 18. Comparison of multitemporal products: (a) optical images, (b) Sentinel-2 10-m LULC Time series, (c) ESA WorldCover, and (d) multitemporal fused bands + FI_NIR$_t$ feature.

exhibited considerable salt-and-pepper noise, making it challenging to distinguish between land classes [Fig. 17(a1)–(a3)]. Using multitemporal optical bands for classification also led to misclassifications between cropland and water bodies, along with cropland and grassland, owing to the resemblance between cropland and water bodies in optical imagery [Fig. 17(b1)–(b3)]. In contrast, using multitemporal fused bands for classification effectively improved misclassification issues between cropland and other land classes, such as water bodies and grassland, providing clearer boundaries between land classes. However, there are still minor misclassifications between cropland and grassland, which may be due to the similar growth patterns between crops and certain types of grassland [Fig. 17(c1)–(c3)]. Compared to the previous three groups, the classification using multitemporal fused bands plus FI_NIR$_t$ feature combines the rich spatial and spectral information from the fused bands with temporal information from the FI_NIR$_t$ feature that captures key phenological changes during the crop growth cycle, so it exhibits the most distinct land class boundaries, with minimal salt-and-pepper noise, and achieves the best differentiation between cropland and other classes such as water bodies and grassland [Fig. 17(d1)–(d3)].

Table VIII presents the quantitative evaluation results for time series classification. Schemes 1–4 denote different combinations of multitemporal polarized channels, multitemporal optical bands, multitemporal fused bands, and multitemporal fused bands plus the FI_NIR$_t$ feature, respectively. The combination of multitemporal fused bands and the FI_NIR$_t$ feature achieves the highest average $F1$, OA, and Kappa values at 93.95%, 95.75%, and 0.9436, respectively. This combination consistently yields the best or subbest $F1$ scores across most ground object types, with notable improvements in accuracy of identifying cropland, buildings, reed, and grassland. However, due to the similarities between grassland and cropland, the accuracy of grassland identification remains relatively lower across the various combinations. In summary, the combination of multitemporal fused bands and the FI_NIR$_t$ feature effectively enhances recognition accuracy across various land classes, thereby facilitating improved differentiation among them.

3) Comparison of Multitemporal Products: The Sentinel-2 10 m Land Use/Land Cover (LULC) Time Series (SS) product provides a global map of LULC derived from ESA Sentinel-2 imagery at a 10-m resolution. The WorldCover product, another 10-m land cover classification product, was generated by using Sentinel-1 and Sentinel-2 images spanning the entire year of 2021, boasting high classification accuracy [50]. We compare the classification results of multitemporal fused bands and the FI_NIR$_t$ feature with these two products, as shown in Fig. 18. The SS product uses a consistent color scheme across different ground object categories as the WorldCover product. The SS product does exhibit significant misclassification issues, particularly between cropland and water bodies, as well as grassland and reed. Similarly, the WorldCover product displays notable misclassification, particularly between cropland and reed. Despite these issues, the WorldCover product offers more accurate classification results than the SS product, attributed to its incorporation of Sentinel-1 imagery. In contrast, our classification results demonstrate notable improvements in addressing these issues,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

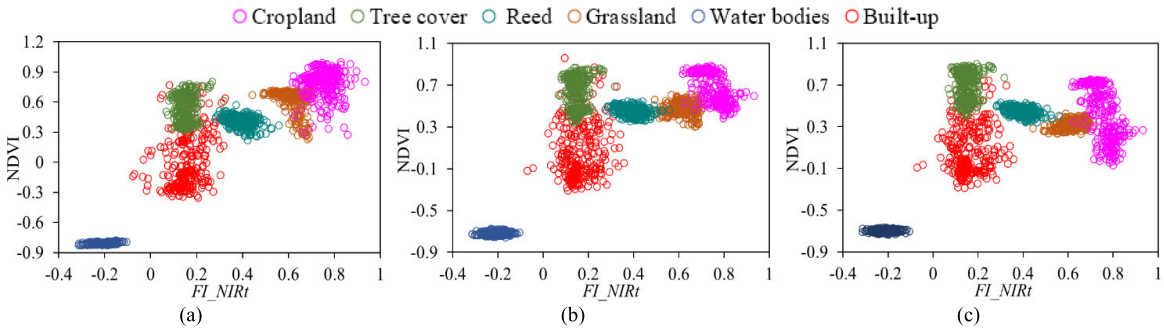


Fig. 19. Sample distribution map of NDVI and FI_NIR$_t$ in (a) August, (b) September, and (c) October.

particularly in accurately differentiating between cropland and water bodies, grassland and reed, as well as cropland and reed.

4) Utility Analysis of Temporal Phenology Feature:

a) Analysis of sample separability effect: Fig. 19 shows the distribution maps of NDVI and FI_NIR$_t$ feature for different ground object samples in August, September, and October. Analysis of Fig. 19(a) reveals that NDVI distinguishes water bodies and buildings effectively from other land classes. However, there is significant overlap among cropland, forest, reed, and grassland samples, rendering them indistinguishable. This suggests that by August, there will be misclassification issues among various vegetation types based on NDVI. In contrast, the FI_NIR$_t$ feature effectively separates water bodies, cropland, reed, grassland, and most other land classes effectively. Overall, the FI_NIR$_t$ feature offers superior differentiation among different land classes compared with NDVI in August. Fig. 19(b) and (c) depicts sample distribution maps for NDVI and FI_NIR$_t$ feature in September and October, revealing trends similar to those in August. In October, there is some misclassification between cropland and buildings, possibly due to ground changes resulting from field harvesting. In summary, across August, September, and October, the FI_NIR$_t$ feature demonstrates superior separability among different land cover classes compared with NDVI, indicating its enhanced ability to differentiate most land cover types.

b) Analysis of constructed feature on classification results: The effects of temporal phenology feature are analyzed from the aspects of single-temporal and multitemporal mapping. The single-temporal mapping is evaluated. In August, incorporating the FI_NIR$_t$ feature yields improvements of 1.35% in Avg.$F1$, 0.9% in OA, and 0.0122 in Kappa. In September, incorporating the FI_NIR$_t$ feature effectively improves the misclassification issues between cropland and reed, leading to increases of 1.43% in Avg.$F1$, 1.79% in OA, and 0.0234 in Kappa. Similarly, in October, incorporating the FI_NIR$_t$ feature reduces salt-and-pepper noise and minimizes misclassification between cropland and reed, resulting in improvements of 2.35% in Avg.$F1$, 1.71% in OA, and 0.0230 in Kappa.

The multitemporal mapping is analyzed. As illustrated in Fig. 17, the incorporation of the FI_NIR$_t$ feature with multitemporal fused bands improves cropland mapping accuracy and reduces misclassification among cropland, grassland, water bodies, buildings, and other land classes. The addition of the FI_NIR$_t$ feature led to an increase in Avg.$F1$, OA, and Kappa scores by 4.59%, 2.02%, and 0.0272, respectively, demonstrating its effectiveness in enhancing the accuracy of cropland mapping.

In summary, the above analysis indicates that the FI_NIR$_t$ feature, derived from fused images during crop heading or flowering stages and the early sowing period, effectively reduces misclassification between cropland and other land classes while enhancing cropland mapping accuracy.

D. Application and Analysis of Various Scenarios

To further validate the effectiveness of the fused bands and FI_NIR$_t$ feature, we selected various different scenarios for analysis, including a total of 12 agricultural regions in China, Canada, USA, and France. The central coordinates for each region are listed in Fig. 20. The NDVI curves revealed that some regions are characterized by single cropping (regions C, D, H, I, J, K, and L), while others practice double cropping (regions A, B, E, F, and G). The optical images show crops growing at the heading or flowering stage. We obtained fused images and FI_NIR$_t$ feature for one growth cycle in each region and conducted classification using samples selected from high-resolution images on Google Earth. By comparing our results with the WorldCover product, we evaluated the recognition performance of the combination of fused images and the FI_NIR$_t$ feature in identifying ground objects.

As shown in Fig. 20, the combination of fused images and the FI_NIR$_t$ feature provides more accurate ground object classification compared with the WorldCover product. It preserves the integrity of ground objects, reduces salt-and-pepper noise, and addresses misclassification issues in cropland, grassland, and woodland areas. In regions A, C, and K, the WorldCover product mistakenly classifies reed areas as cropland, whereas the combination of fused images and FI_NIR$_t$ feature corrects this error. In regions B, E, F, G, I, and L, the WorldCover product exhibits significant misclassification, particularly between cropland, grassland, and woodland. This issue is most pronounced in regions E, I, and L, where extensive misclassification occurs between grassland and woodland, as well as between grassland and cropland. In contrast, our approach yields better differentiation of land classes in these regions, closely matching the real ground distribution. Regions D and H show close alignment between the WorldCover product and our results, likely due to simpler land types and fewer mixed distributions of cropland, woodland, reed, and other land types. In summary,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

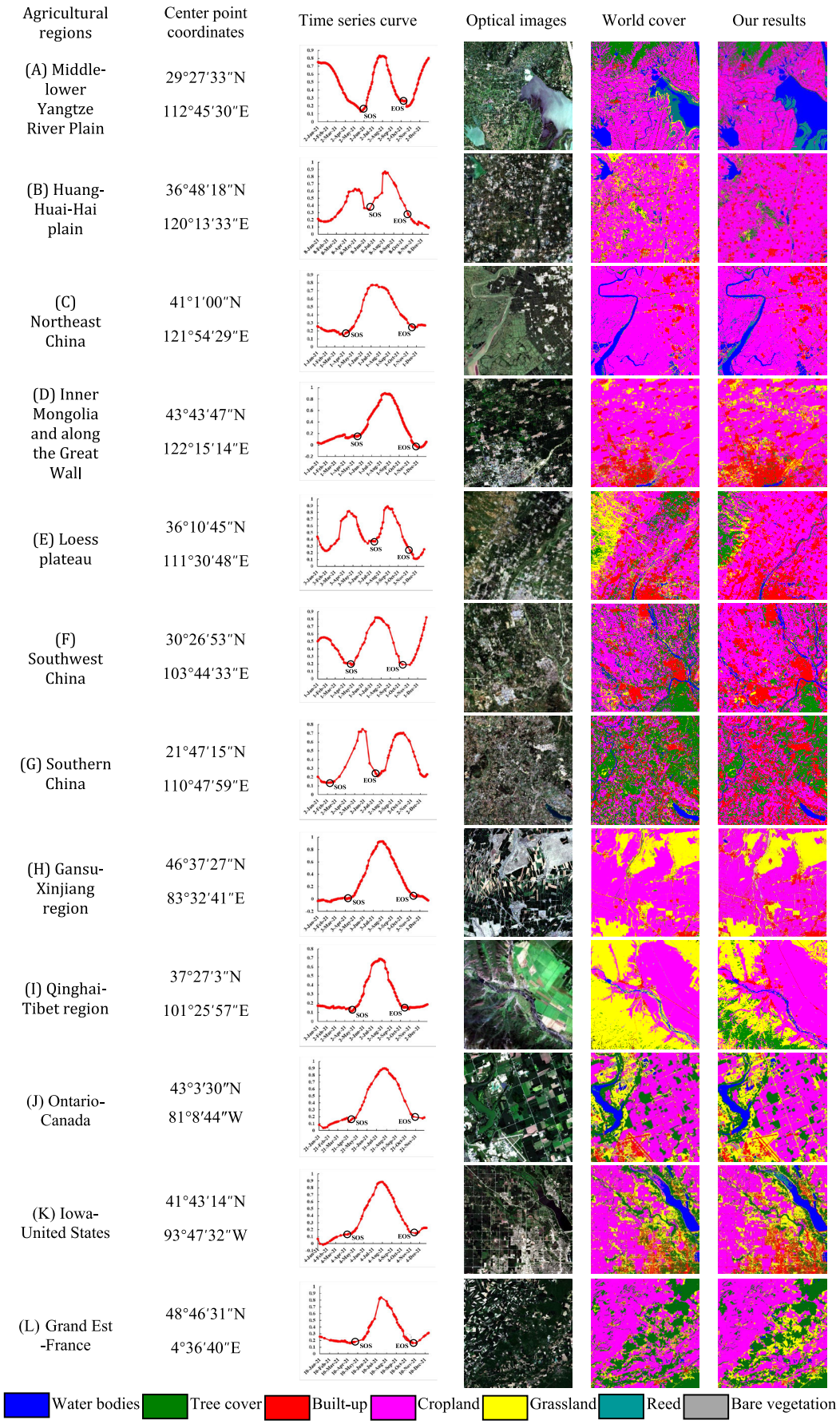


Fig. 20. Classification results for 12 agricultural regions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

the combination of fused images and the FI_NIR$_t$ feature enhances ground object classification, reduces salt-and-pepper noise, and improves classification accuracy, demonstrating its effectiveness for land classification applications.

IV. Conclusion

This article proposes a novel Sentinel-1 and Sentinel-2 hybrid method, which combines PCA transform with CSR and GAN fusion strategies in NSST domain to produce fused images with rich spatial and spectral information. Subsequently, we construct temporal phenology feature based on the change rules of ground objects on the fused images throughout the crop growth cycle to effectively distinguish crops from low woodland and improve the accuracy of cropland mapping.

Experimental results demonstrate the superiority of the proposed fusion method in retaining spatial information and reducing spectral distortion. It outperforms six classical methods and five advanced deep learning methods in both qualitative and quantitative assessments. Using the RF algorithm for classification reveals that the combination of fused images and temporal phenology feature achieves the highest overall accuracies in August (92.26%), September (88.65%), and October (90.33%) compared to independent single-temporal optical and fused images. Similarly, the combination also achieves the highest OA at 95.75% compared to the independent multitemporal polarized, optical, and fused images. In addition, the fused images and temporal phenology feature are tested in a total of 12 agricultural regions in China, Canada, USA, and France and compared to the well-classified WorldCover product, further proving their effectiveness in recognizing ground objects.

Future research aims to develop new fusion methods using other deep learning models and build fusion models based on polarimetric target decomposition. Furthermore, new phenology features will be explored by delving into the characteristics of additional RE bands throughout the crop growth cycle.

Code Availability

The code generated and analyzed during the current study is available upon request from the corresponding author. We encourage interested parties to contact us via email to obtain a copy of the code.

References

[1] J. Cheng, D. Xiang, Q. Yin, and F. Zhang, "A novel crop classification method based on the tensor-GCN for time-series PolSAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5237614.

[2] S. Yang, L. Gu, X. Li, F. Gao, and T. Jiang, "Fully automated classification method for crops based on spatiotemporal deep-learning fusion technology," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405016.

[3] B. Wu et al., "Challenges and opportunities in remote sensing-based crop monitoring: A review," *Nat. Sci. Rev.*, vol. 10, no. 4, Mar. 2023, Art. no. nwac29.

[4] Y. Xu, H. Ebrahimy, and Z. Zhang, "Bayesian joint adaptation network for crop mapping in the absence of mapping year ground-truth samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4412220.

[5] D. Ienco, R. Interdonato, R. Gaetano, and D. Ho Tong Minh, "Combining Sentinel-1 and Sentinel-2 satellite image time series for land cover mapping via a multi-source deep learning architecture," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 11–22, Dec. 2019.

[6] C. Zhou, J. Huang, Y. Xiao, M. Du, and S. Li, "A novel approach: Coupling prior knowledge and deep learning methods for large-scale plastic greenhouse extraction using Sentinel-1/2 data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104073.

[7] T. Liu, P. Li, F. Zhao, J. Liu, and R. Meng, "Early-stage mapping of winter canola by combining Sentinel-1 and Sentinel-2 data in Jianghan Plain China," *Remote Sens.*, vol. 16, no. 17, p. 3197, Aug. 2024.

[8] H. Li et al., "Automated in-season crop-type data layer mapping without ground truth for the conterminous United States based on multisource satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4403214.

[9] W. Li et al., "Aligning semantic distribution in fusing optical and SAR images for land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 272–288, May 2023.

[10] H. Ghassemian, "A review of remote sensing image fusion methods," *Inf. Fusion*, vol. 32, pp. 75–89, Nov. 2016.

[11] Y. Ye, J. Zhang, L. Zhou, J. Li, X. Ren, and J. Fan, "Optical and SAR image fusion based on complementary feature decomposition and visual saliency features," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5205315.

[12] L. Xu et al., "Parcel level staple crop type identification based on newly defined red-edge vegetation indices and ORNN," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 108012.

[13] J. Adrian, V. Sagan, and M. Maimaitijiang, "Sentinel SAR-optical fusion for crop type mapping using deep learning and Google Earth engine," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 215–235, May 2021.

[14] X. Liu, H. Zou, S. Wang, Y. Lin, and X. Zuo, "Joint network combining dual-attention fusion modality and two specific modalities for land cover classification using optical and SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3236–3250, 2024.

[15] S. C. Kulkarni and P. P. Rege, "Pixel level fusion techniques for SAR and optical images: A review," *Inf. Fusion*, vol. 59, pp. 13–29, Jul. 2020.

[16] Y. Liu, X. Chen, H. Peng, and Z. Wang, "Multi-focus image fusion with a deep convolutional neural network," *Inf. Fusion*, vol. 36, pp. 191–207, Jul. 2017.

[17] C. Sui et al., "IG-GAN: Interactive guided generative adversarial networks for multimodal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634719.

[18] H. Zhang, Z. Le, Z. Shao, H. Xu, and J. Ma, "MFF-GAN: An unsupervised generative adversarial network with adaptive and gradient joint constraints for multi-focus image fusion," *Inf. Fusion*, vol. 66, pp. 40–53, Feb. 2021.

[19] Q. Chen, M. Pang, X. Liu, and Z. Zhang, "A polarization-spectrum fusion framework based on multiscale transform and generative adversarial network for improving water and different vegetation distinguishability," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 123, Sep. 2023, Art. no. 103468.

[20] Y. Liu, X. Chen, Z. Wang, Z. J. Wang, R. K. Ward, and X. Wang, "Deep learning for pixel-level image fusion: Recent advances and future prospects," *Inf. Fusion*, vol. 42, pp. 158–173, Jul. 2018.

[21] Z. Zhang et al., "Fusion of infrared and visible images via multi-layer convolutional sparse representation," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 6, Jul. 2024, Art. no. 102090.

[22] F. Xu, Y. Shi, W. Yang, G.-S. Xia, and X. X. Zhu, "CloudSeg: A multimodal learning framework for robust land cover mapping under cloudy conditions," *ISPRS J. Photogramm. Remote Sens.*, vol. 214, pp. 21–32, Aug. 2024.

[23] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[24] X. Huang, J. Wang, J. Shang, C. Liao, and J. Liu, "Application of polarization signature to land cover scattering mechanism analysis and classification using multi-temporal C-band polarimetric RADARSAT-2 imagery," *Remote Sens. Environ.*, vol. 193, pp. 11–28, May 2017.

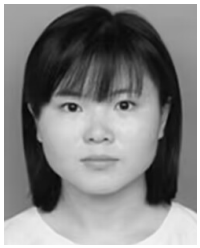
[25] A. Veloso et al., "Understanding the temporal behavior of crops using Sentinel-1 and Sentinel-2-like data for agricultural applications," *Remote Sens. Environ.*, vol. 199, pp. 415–426, Sep. 2017.

[26] L. Fan et al., "A temporal-spatial deep learning network for winter wheat mapping using time-series Sentinel-2 imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 214, pp. 48–64, Aug. 2024.

[27] M. G. Maponya, A. van Niekerk, and Z. E. Mashimbye, "Pre-harvest classification of crop types using a Sentinel-2 time-series and machine learning," *Comput. Electron. Agricult.*, vol. 169, Feb. 2020, Art. no. 105164.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.

[28] F. Guo et al., "Multi-dimensional fusion of spectral and polarimetric images followed by pseudo-color algorithm integration and mapping in HSI space," *Remote Sens.*, vol. 16, no. 7, p. 1119, Mar. 2024.
[29] Y. Quan, R. Zhang, J. Li, S. Ji, H. Guo, and A. Yu, "Learning SAR-optical cross modal features for land cover classification," *Remote Sens.*, vol. 16, no. 2, p. 431, Jan. 2024.
[30] G. Easley, D. Labate, and W.-Q. Lim, "Sparse directional image representations using the discrete shearlet transform," *Appl. Comput. Harmon. Anal.*, vol. 25, no. 1, pp. 25–46, Jul. 2008.
[31] J. Xu, D. Hong, S. Su, R. Fang, and H. Li, "Infrared and visible image fusion based on saliency detection dynamic threshold neural P systems in non-subsampled shearlet transform domain," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.
[32] Y. Liu, X. Chen, R. K. Ward, and Z. J. Wang, "Image fusion with convolutional sparse representation," *IEEE Signal Process. Lett.*, vol. 23, no. 12, pp. 1882–1886, Dec. 2016.
[33] B. Wohlberg, "Efficient algorithms for convolutional sparse representations," *IEEE Trans. Image Process.*, vol. 25, no. 1, pp. 301–315, Jan. 2016.
[34] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708.
[35] S. Pal, T. Majumdar, and A. K. Bhattacharya, "ERS-2 SAR and IRS-1C LISS III data fusion: A PCA approach to improve remote sensing based geological interpretation," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 5, pp. 281–297, 2007.
[36] S. A. Manaf, N. Mustapha, M. N. Sulaiman, N. A. Husin, and M. R. A. Hamid, "Comparison of classification techniques on fused optical and SAR images for shoreline extraction: A case study at northeast coast of peninsular Malaysia," *J. Comput. Sci.*, vol. 12, no. 8, pp. 399–411, Aug. 2016.
[37] M. Yin, X. Liu, Y. Liu, and X. Chen, "Medical image fusion with parameter-adaptive pulse coupled neural network in nonsubsampled shearlet transform domain," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 1, pp. 49–64, Jan. 2018.
[38] Y. Liu, S. Liu, and Z. Wang, "A general framework for image fusion based on multi-scale transform and sparse representation," *Inf. Fusion*, vol. 24, pp. 147–164, Jul. 2015.
[39] J. Ma, W. Yu, P. Liang, C. Li, and J. Jiang, "FusionGAN: A generative adversarial network for infrared and visible image fusion," *Inf. Fusion*, vol. 48, pp. 11–26, Aug. 2019.
[40] W. Tang, F. He, and Y. Liu, "YDTR: Infrared and visible image fusion via Y-shape dynamic transformer," *IEEE Trans. Multimedia*, vol. 25, pp. 5413–5428, 2022.
[41] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via Swin transformer," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.
[42] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5906–5916.
[43] L. Alparone, S. Baronti, A. Garzelli, and F. Nencini, "A global quality measurement of pan-sharpened multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 313–317, Oct. 2004.
[44] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Aug. 2002.
[45] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, and L. M. Bruce, "Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data-fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3012–3021, Oct. 2007.
[46] Y.-J. Rao, "In-fibre Bragg grating sensors," *Meas. Sci. Technol.*, vol. 8, no. 4, p. 355, 1997.
[47] A. M. Eskicioglu and P. S. Fisher, "Image quality measures and their performance," *IEEE Trans. Commun.*, vol. 43, no. 12, pp. 2959–2965, Dec. 1995.
[48] P. Balasubramaniam and V. P. Ananthi, "Image fusion using intuitionistic fuzzy sets," *Inf. Fusion*, vol. 20, pp. 21–30, Nov. 2014.
[49] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.
[50] D. Zanaga et al., "ESA WorldCover 10 m 2021 v200," Tech. Rep., 2022.



Mengqing Pang (Graduate Student Member, IEEE) received the B.E. degree in remote sensing science and technology from Henan Polytechnic University, Jiaozuo, China, in 2021.

She is engaged in SAR and optical image fusion and application.



Qihao Chen (Member, IEEE) received the B.E. degree in geographic information system and the M.E. and Ph.D. degrees in cartography and geographic information engineering from China University of Geosciences (CUG), Wuhan, China, in 2004, 2007, and 2011, respectively.

He is currently an Associate Professor with the School of Geography and Information Engineering, CUG. His research interests include polarimetric synthetic aperture radar information extraction and application.



Jiali Shang received the B.Sc. degree in geography from Beijing Normal University, Beijing, China, in 1984, the M.A. degree from the University of Windsor, Windsor, ON, Canada, in 1996, and the Ph.D. degree in remote sensing from the University of Waterloo, Waterloo, ON, Canada, in 2005.

She is currently a Research Scientist with the Science and Technology Branch of Agriculture and Agri-Food Canada, Ottawa, ON, Canada. She is involved in the methodology development and application of Earth observation (EO) technology to vegetation biophysical parameter retrieval, crop growth modeling, precision agriculture, and the detection of field operation activity related to crop production using both optical and radar EO data.



Weijia Long received the B.E. degree in surveying and mapping engineering from the Central South University of Forestry and Technology (CSUFT), Changsha, China, in 2022.

His research interests include ecological restoration and remote sensing.



Xiuguo Liu (Member, IEEE) received the B.E. and M.E. degrees in computer and application and the Ph.D. degree in cartography and geographic information engineering from China University of Geosciences (CUG), Wuhan, China, in 1991, 1999, and 2004, respectively.

He is currently a Professor with the School of Geography and Information Engineering, CUG. His research interests include polarimetric synthetic aperture radar theories, geographic information utilization, and 3-D geological modeling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:40:33 UTC from IEEE Xplore. Restrictions apply.